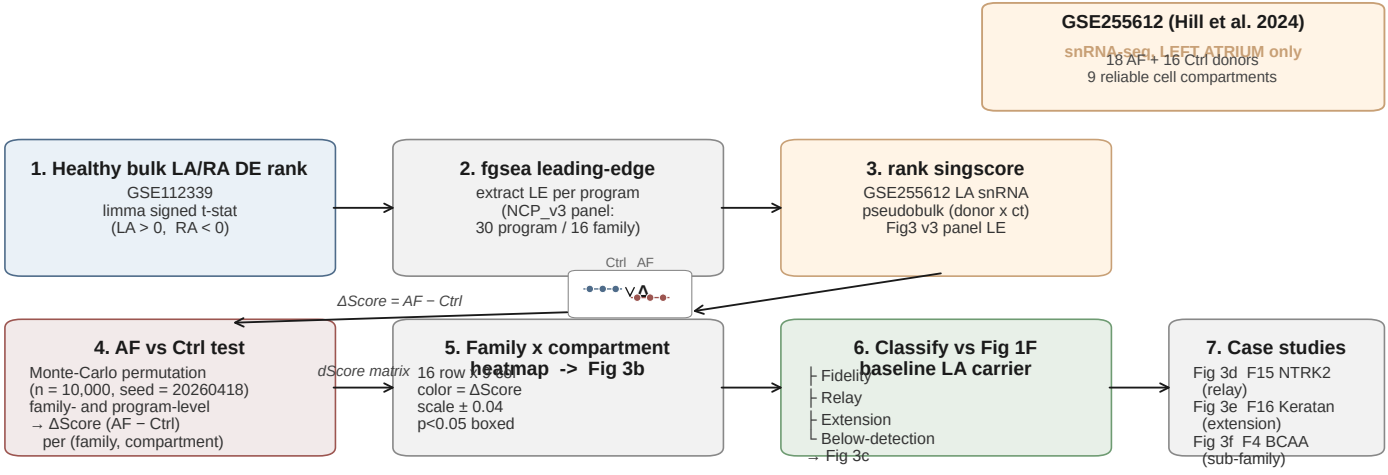
## Fig 3a AF cellular substrate validation pipeline (LA snRNA)




**Methodological limitations (acknowledged in Discussion)**

(i) GSE255612 contains LEFT ATRIUM samples only; RA-side cellular activity changes are not observable in this dataset.

(ii) Compartment-level pseudobulk dilutes within-cell-type rare-population signals; bulk rewiring may reside below per-compartment detection.

(iii) Cellular relay vs sub-cell-type composition shift cannot be fully separated at the pseudobulk level (single-nucleus subtype analysis is reserved as future work).

(iv) Fig 3 baseline carrier reference comes from the HCA paired LA + RA atlas (Fig 1F); the AF-time activity-shift cell type is determined within GSE255612 LA arm only.